

Case Study #2: Forecasting Walmart's Revenue

The data set for case study #2 represents quarterly revenues (in \$million) in Walmart from the first quarter of 2006 through the third quarter (Q3) of 2025 (*673_case2.csv*). This quarterly data is collected from www.macrotrends.net/stocks/charts/WMT/walmart/revenue. The goal is to forecast Walmart's quarterly revenue in Q4 of 2025 and four quarters (Q1-Q4) of 2026-2027.

Questions

1. Plot the data and visualize time series components.

- Create time series data set in R using the *ts()* function.
- Apply the *plot()* function to create a data plot with historical data, provide it in your report, and explain what time series components can be visualized in this plot.

2. Apply five regression models using data partition.

Consider the following 3 regression-based models:

- Regression model with seasonality
 - Regression model with linear trend and seasonality
 - Regression model with quadratic trend and seasonality.
- Develop data partition with the validation partition of 19 periods and the rest for the training partition.
 - Use the *tslm()* function for the training partition to develop each of the 3 regression models from the above list. Apply the *summary()* function to identify the model structure and parameters for each regression model, show them in your report, and also present the respective model equation and define its predictors. Briefly explain if the model is a good fit, statistically significant, and thus may be applied for forecasting. Use each model to forecast revenues for the validation period using the *forecast()* function, and present this forecast in your report.
 - Apply the *accuracy()* function to compare performance measures of the 3 regression models you developed in 2b. Present the accuracy measures in your report, compare them, and, using MAPE and RMSE, identify the *two most accurate regression models for forecasting*.

3. Employ the entire data set to make time series forecast.

- Apply the two most accurate regression models identified in 2c to make the forecast in Q4 of 2025 and the four quarters (Q1-Q4) of 2026-2027. For that, use the entire data set to develop the regression model using the *tslm()* function. Apply the *summary()* function to identify the model structure and parameters, show them in your report, and also present the respective model equation and define its predictors. Briefly explain if the model is a good fit, statistically significant, and thus may be applied for forecasting. Use each model to forecast Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027 using the *forecast()* function, and present this forecast in your report.
- Apply the *accuracy()* function to compare the performance measures of the regression models developed in 3a with those for the naïve and seasonal naïve forecasts. Present the accuracy measures in your report, compare them, and identify, using MAPE and RMSE, which forecast is most accurate to forecast Walmart's quarterly revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.